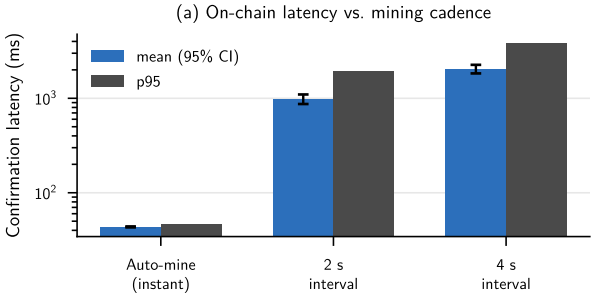
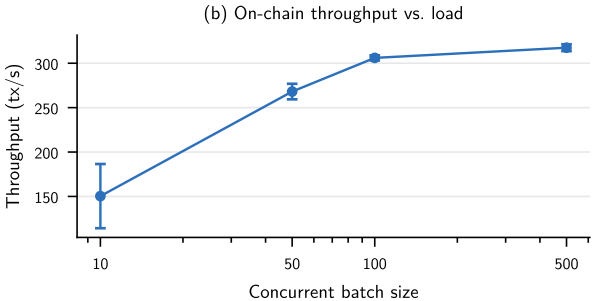
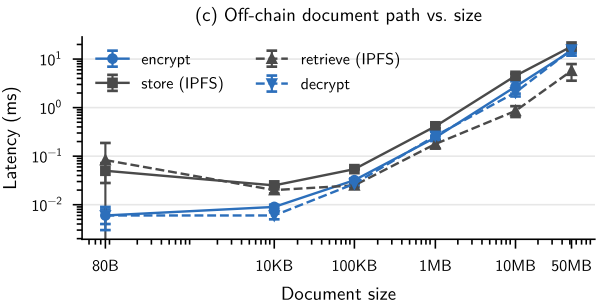
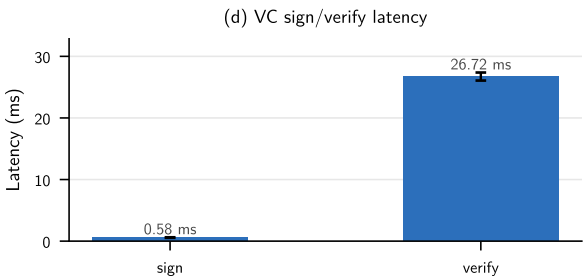

(a) On-chain latency vs. mining cadence
Confirmation latency (ms)
mean (95% CI)
p95
10^3
10^2
Auto-mine (instant)
2 s interval
4 s interval


(b) On-chain throughput vs. load
Throughput (tx/s)
300
250
200
150
10
50
100
500
Concurrent batch size


(c) Off-chain document path vs. size
Latency (ms)
encrypt
store (IPFS)
retrieve (IPFS)
decrypt
10^1
10^0
10^{-1}
10^{-2}
80B
10KB
100KB
1MB
10MB
50MB
Document size


(d) VC sign/verify latency
Latency (ms)
30
20
10
0
0.58 ms
26.72 ms
sign
verify